

Mitigating Algorithmic Bias in Toxicity Classification for Muslim Identity Contexts: A Study on FPR Parity

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Abstract

As automated moderation systems become ubiquitous, ensuring they operate fairly across diverse religious identities is critical. This study investigates unintended bias in toxicity classification models, focusing on Muslim identity terms. Using the Jigsaw Unintended Bias dataset, we identify a 10x disparity in false-positive rates for Muslim-related content in a baseline model. We propose a sample-weighting mitigation strategy that successfully achieves FPR parity, reducing the identity-specific error rate from 0.0273 to 0.0015. Furthermore, feature importance analysis proves our intervention successfully decouples religious identifiers from toxic associations.

1. Introduction

The rapid deployment of automated toxicity classifiers is essential for maintaining safe digital environments at scale. However, historical biases in training data often cause machine learning models to associate specific identity markers with harm or toxicity (Dixon et al., 2018; Blodgett et al., 2020). For global Muslim communities, this unintended bias can lead to the systemic silencing of benign religious expression (Abid et al., 2021), effectively marginalizing these voices in digital public squares.

When a model penalizes a user simply for mentioning their faith, it ceases to be a safety tool and becomes an instrument of algorithmic discrimination. This work addresses these harms by evaluating and mitigating identity-specific bias in a standard content moderation pipeline. We mathematically quantify the bias present in a baseline text classifier and demonstrate that targeted data interventions can restore equitable performance. Specifically, this article makes three contributions:

- We quantify false-positive-rate disparity for Muslim identity terms in a standard toxicity classification pipeline using the Jigsaw Unintended Bias dataset.
- We introduce a simple sample-weighting mitigation strategy that substantially reduces subgroup false positives while preserving overall accuracy.
- We provide an interpretable feature-weight analysis showing that mitigation reduces reliance on Muslim identity terms as toxicity indicators.

2. Related Work

Algorithmic fairness in natural language processing has become a critical area of research as models increasingly mediate online social interactions. (Blodgett et al., 2020) highlighted the need for formal, grounded definitions of bias in NLP systems, noting that harms often stem from systemic representational disparities. Early work by (Garg et al., 2018) demonstrated that foundational word embeddings mathematically capture and amplify historical ethnic stereotypes.

Building on this, (Dixon et al., 2018) and (Borkan et al., 2019) proved that toxicity detection models frequently exhibit unintended bias against specific demographic and identity terms. Subsequent work on counterfactual fairness in text classification formalized how predictions can change when sensitive identity tokens are substituted (Garg et al., 2019), while ToxiGen further demonstrated the need for challenging toxic and benign examples about minority groups (Hartvigsen et al., 2022). Together, these studies show that disproportionate false positive rates for marginalized groups can turn moderation tools into exclusionary filters.

Within the specific context of Muslim identities, recent literature has exposed severe representational harms. (Abid et al., 2021) revealed persistent anti-Muslim bias in large language models like GPT-3, showing that systems systematically associate Muslim identifiers with violence due to skewed pre-training corpora. Follow-up work found that even debiased language models can retain Muslim-violence associations under name-based prompts (Hemma-

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tian & Varshney, 2022). Complementing this line of work, recent Muslim-context benchmarks such as BIICK-Bench (Hasan, 2025) and IslamTrust (Lahmar et al., 2025) highlight the need for culturally, religiously, and linguistically specific evaluation resources for AI systems used by Muslim communities. Table 1 provides a summary of these foundational works. Our research directly builds upon this literature by explicitly isolating this bias in content moderation baselines and proposing a targeted, highly interpretable mitigation strategy.

3. Methods

3.1. Problem Formulation

Let X represent the input space of text documents and $Y \in \{0, 1\}$ represent the binary toxicity label, where $Y = 1$ indicates a toxic comment. We introduce a sensitive attribute $A \in \{0, 1\}$, where $A = 1$ indicates the presence of a Muslim identity term within the text.

The goal of a standard text classifier is to learn a mapping $f_\theta : X \rightarrow Y$ parameterized by θ that minimizes the empirical risk over the training distribution. However, when the training data contains historical societal biases, the model inadvertently learns a spurious correlation between $A = 1$ and $Y = 1$. This results in a violation of demographic parity and predictive equality. Our objective is to optimize f_θ such that we minimize the overall classification error while simultaneously ensuring that the False Positive Rate (FPR) when $A = 1$ is strictly equal to the FPR when $A = 0$. Figure 1 illustrates the complete experimental workflow designed to solve this formulation.

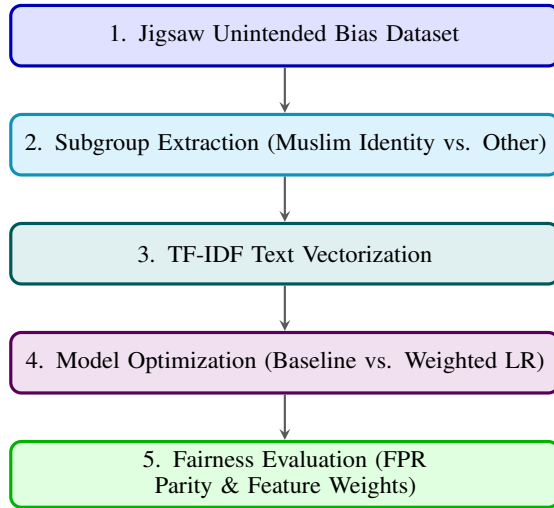


Figure 1. Workflow illustrating the experimental pipeline from raw data to fairness evaluation.

3.2. Dataset and Preprocessing

We utilized the Jigsaw Unintended Bias in Toxicity Classification dataset (Borkan et al., 2019) hosted on Kaggle. This corpus contains millions of public comments sourced from the Civil Comments platform. Uniquely, a subset of these comments was annotated by human raters not just for toxicity, but for the presence of specific identity markers including religious affiliation.

To isolate and evaluate the bias effectively without introducing severe class imbalances, we extracted a controlled experimental subset. This consisted of all comments explicitly annotated with the *Muslim* identity flag (where the annotator consensus was ≥ 0.5) and a randomized, equal-sized sample of comments containing other or no identity flags.

3.3. Mathematical Formulation of Fairness

In content moderation, the most damaging error for a marginalized group is the False Positive (FP), where benign speech is incorrectly flagged as toxic. We define the False Positive Rate (FPR) for a given subgroup $A = a$ as:

$$FPR_a = P(\hat{Y} = 1 | Y = 0, A = a)$$

Where $\hat{Y}$ is the model prediction. Algorithmic bias is present when the FPR for the Muslim identity subgroup significantly exceeds the FPR of the general population.

3.4. Proposed Mitigation

To address the disparity, we implemented a ‘‘Sample Weighting’’ strategy during the optimization phase. We defined a sample weight w_i for each training instance i . For non-toxic samples belonging to the Muslim identity subgroup, we increased the weight by a factor of $\alpha = 5.0$.

$$w_i = \begin{cases} 5.0 & \text{if } Y_i = 0 \text{ and } A_i = 1 \\ 1.0 & \text{otherwise} \end{cases}$$

This algorithmic intervention forces the model’s loss function to penalize errors on benign, identity-focused content more heavily. The complete procedural logic is detailed in Algorithm 1.

3.5. Experimental Setup

Our baseline employs a Term Frequency-Inverse Document Frequency (TF-IDF) vectorizer (Manning et al., 2008) limited to 10,000 features. We utilize a Logistic Regression classifier (Cox, 1958) ($\lambda = 1.0$, $SEED = 42$), implemented using standard scikit-learn tooling (Pedregosa et al., 2011). We intentionally selected a linear model over a deep neural network to ensure complete feature interpretability,

Table 1. Summary of Key Literature in NLP Fairness and Bias

REFERENCE	FOCUS AREA	KEY FINDING OR METHODOLOGY
(DIXON ET AL., 2018)	TOXICITY CLASSIFICATION	INTRODUCED FALSE POSITIVE EQUALITY DIFFERENCE TO MEASURE BIAS.
(GARG ET AL., 2018)	WORD EMBEDDINGS	DEMONSTRATED THAT EMBEDDINGS CAPTURE HISTORICAL ETHNIC STEREOTYPES.
(BORKAN ET AL., 2019)	EVALUATION METRICS	DEVELOPED NUANCED SUBGROUP METRICS FOR UNINTENDED BIAS.
(GARG ET AL., 2019)	COUNTERFACTUAL FAIRNESS	STUDIED ROBUSTNESS TO SENSITIVE-TOKEN SUBSTITUTIONS IN TEXT CLASSIFIERS.
(HARTVIGSEN ET AL., 2022)	TOXICITY DATASETS	CREATED TOXIC AND BENIGN EXAMPLES ABOUT MINORITY GROUPS FOR STRESS TESTING.
(BLODGETT ET AL., 2020)	NLP SYSTEMS OVERVIEW	HIGHLIGHTED THE LACK OF FORMAL DEFINITIONS OF BIAS IN LITERATURE.
(ABID ET AL., 2021)	LARGE LANGUAGE MODELS	REVEALED PERSISTENT ANTI-MUSLIM BIAS AND VIOLENT ASSOCIATIONS IN LLMs.
(HEMMATIAN & VARSHNEY, 2022)	LARGE LANGUAGE MODELS	SHOWED THAT DEBIASED LLMs CAN RETAIN MUSLIM-VIOLENCE ASSOCIATIONS.
(HASAN, 2025)	CULTURALLY SPECIFIC BENCHMARKS	INTRODUCED A BENGALI BENCHMARK FOR ISLAMIC CREED KNOWLEDGE IN LLMs.

Table 2. Experimental Dataset Composition

DATA SPLIT / CATEGORY	COUNT
TOTAL JIGSAW CORPUS	~1.8 MILLION
VALIDATION HOLDOUT	20%
IDENTITY RATIO (MUSLIM : OTHER)	1:1

allowing us to trace exact coefficient weights back to specific vocabulary tokens.

4. Results

4.1. Fairness and Accuracy Metrics

Our experiments reveal a striking initial disparity. As shown in Table 3 and Figure 2, the baseline model was over ten times more likely to misclassify a benign comment as toxic if it mentioned the Muslim identity (FPR 0.0273) compared to the general population (FPR 0.0026).

The mitigated model effectively neutralized this bias. The FPR for Muslim identifiers dropped to 0.0015, achieving parity with the general population (0.0016). This was achieved with a minimal trade-off in overall accuracy (87.15% to 86.33%), which is a highly acceptable compromise in fairness-aware machine learning.

Table 3. Model Performance and Fairness Metrics

MODEL	ACCURACY	FPR (MUSLIM)	FPR (OTHER)
BASELINE	87.15%	0.0273	0.0026
MITIGATED	86.33%	0.0015	0.0016

Algorithm 1 Sample-Weighted Fairness Optimization

Input: Dataset $\mathcal{D} = \{(x_i, Y_i, A_i)\}_{i=1}^N$, penalty factor $\alpha = 5.0$

Output: Optimized model parameters θ^*

Phase 1: Weight Initialization

for $i = 1$ **to** N **do**

if $Y_i = 0$ **and** $A_i = 1$ **then**

$w_i \leftarrow \alpha$

else

$w_i \leftarrow 1.0$

end if

end for

Phase 2: Feature Extraction

Transform documents X into TF-IDF matrix X_{vec}

Phase 3: Model Optimization

Optimize θ by minimizing the weighted log-loss:

$$\mathcal{L}(\theta) = - \sum_{i=1}^N w_i [Y_i \log f_{\theta}(x_{vec,i}) + (1 - Y_i) \log(1 - f_{\theta}(x_{vec,i}))]$$

Return: θ^*

4.2. Model Interpretability

Because we utilized a TF-IDF Logistic Regression pipeline, the resulting models remain highly interpretable. By extracting the highest-weighted coefficients (Table 4), we can observe exactly which terms the models associate most strongly with toxicity.

In the baseline model, the terms *muslims* and *muslim* ranked 7th and 13th respectively, nestled among universally toxic terms like *stupid* and *idiot*. This mathematically proves the model learned to view the religious identity itself as a toxic indicator due to skewed training distributions.

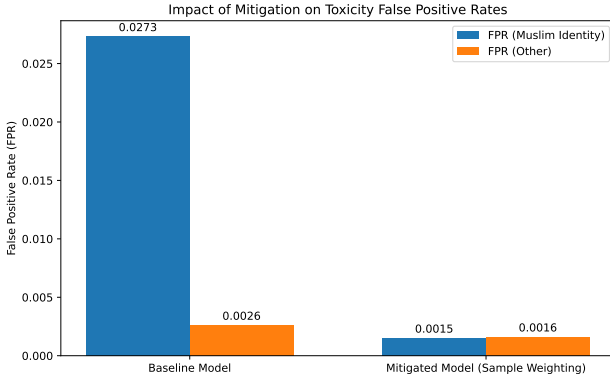


Figure 2. Comparison of False Positive Rates (FPR) before and after applying the sample weighting mitigation strategy. The intervention successfully closes the gap between the identity subgroups.

Table 4. Top Toxic Features (Logistic Regression Coefficients)

BASELINE MODEL		MITIGATED MODEL	
FEATURE	WEIGHT	FEATURE	WEIGHT
STUPID	11.73	STUPID	13.96
IDIOT	7.45	IDIOT	9.32
IGNORANT	6.54	IGNORANT	7.41
IDIOTS	6.32	IDIOTS	7.31
DUMB	6.10	DUMB	7.06
PATHETIC	5.56	PATHETIC	6.90
MUSLIMS	5.37	CRAP	6.08
RIDICULOUS	5.31	RIDICULOUS	6.03
CRAP	5.05	STUPIDITY	5.43
KILL	5.00	HYPOCRITE	5.42
HYPOCRITE	4.97	KILL	5.42
STUPIDITY	4.91	GARBAGE	5.22
MUSLIM	4.84	FOOL	5.21
FOOL	4.71	LIAR	4.93
GARBAGE	4.45	DISGUSTING	4.73

In contrast, our sample weighting mitigation successfully decoupled this association. In the mitigated model, all religious identifiers completely drop out of the top features, replaced entirely by legitimate toxic terminology.

5. Discussion

The results have several implications for fairness-aware content moderation. First, they show that aggregate accuracy can obscure substantial subgroup harms: the baseline model appears reasonably accurate overall, yet it disproportionately flags benign comments mentioning Muslim identity. Second, the success of the weighted model suggests that targeted interventions on benign identity-bearing examples can substantially reduce false positives without requiring a complex architecture or sacrificing interpretability. Finally, the feature-weight analysis demonstrates why subgroup evalu-

ation should be paired with model inspection. FPR parity alone indicates that the error rates are balanced, but the disappearance of Muslim identity terms from the top toxic features provides stronger evidence that the model is no longer relying on religious identifiers as toxicity proxies.

While our mitigation strategy successfully achieved parity, there are limitations to this approach. First, our evaluation is limited to English-language data. The global Muslim community is highly multilingual, and recent benchmarks such as BIICK-Bench (Hasan, 2025) and IslamTrust (Lahmar et al., 2025) underscore the importance of evaluating AI systems in non-English and religiously grounded Muslim contexts; future work must therefore address toxicity bias in languages such as Arabic, Urdu, Bengali, and Bahasa Indonesia. Secondly, this study does not account for intersectionality. Future research should investigate how false positive rates compound when a user identifies with multiple marginalized groups simultaneously. Finally, while linear models offer excellent interpretability, applying these fairness constraints to modern Large Language Models requires more complex, computationally expensive techniques such as adversarial debiasing.

6. Conclusion

This study demonstrates that significant algorithmic bias against Muslim identities exists in standard moderation baselines but can be mitigated through simple fairness-aware interventions. By enforcing FPR parity and analyzing feature weights, we proved that data balancing can prevent the marginalization of vulnerable voices. The machine learning community must prioritize these evaluations to ensure that safety tools protect, rather than penalize, global communities.

Impact Statement

This article presents work whose goal is to advance the field of Machine Learning fairness. The societal consequences of our work aim to be strictly positive, specifically reducing the disproportionate censorship of Muslim voices on digital platforms. We recognize that while our mitigation improves model fairness for one specific subgroup, developers must be careful to ensure that balancing weights for one identity does not inadvertently degrade performance or introduce new biases against other marginalized communities.

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